Exploring Question Quality within CQA Platforms

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Intro

- A CQA is a Community Question & Answer Platform (e.g. Stack Overflow, Meta Stack Exchange, etc.)

- We built on a previous work: https://www.kaggle.com/datasets/imoore/60k-stack-overflow-questions-with-quality-rate/data
  - We use the same target classes throughout our project

- We expanded the data set by incorporated other CQAs, in addition to Stack Overflow

Why CQAs

- CQAs are valuable resources for figuring out solutions to problems, and often one of the first places people go after running into an issue (who hasn't visited Stack Overflow before?)

- Many questions are posted on these sites every day, which makes it hard for moderators to keep up with everything

- Creating a classifier that can detect post quality can help moderators to review fewer questions and streamline their work

- It can help question askers get quicker and more helpful answers

Our Project

Our additional dataset was comprised of questions from multiple CQAs

- Gathered data from https://data.stackexchange.com
- The quality splits were:
  - High Quality (HQ)
  - Low Quality Edited (LQ_EDIT)
  - Low Quality Closed (LQ_CLOSE)
- CQAs included were:
  - Ask Ubuntu
  - Mathematics
  - Meta Stack Exchange
  - Server Fault
  - Super User

Our Project

- We started by trying out a handful of models to see how they performed on the original dataset
  - KNN
  - Random Forest
  - Support Vector Classifier
  - Decision Tree
  - Hist Gradient Boosting
  - Logistic Regression
  - MLP Classifier
  - Transformer
- From there we did some hyperparameter tuning and tried an ensemble
  - Random Forest (trees)
  - Hist Gradient Boosting (learning rate and max depth)
  - Logistic Regression + SVC + Random Forest

Results

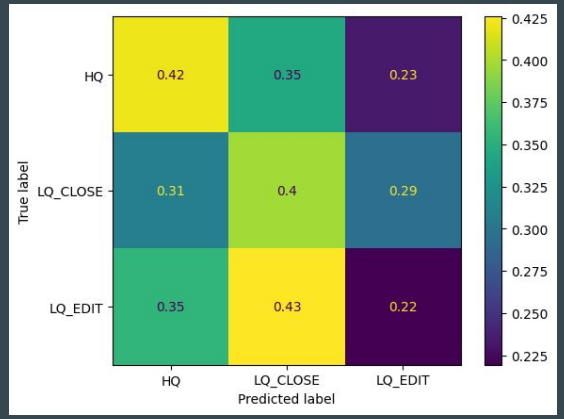
Model	Precision	Recall	F1 Score	AUC-ROC
KNN	0.488	0.363	0.248	0.531
Random Forest	0.761	0.761	0.761	0.907
Naive Bayes	0.71	0.694	0.692	0.872
Support Vector Classifier	0.773	0.775	0.773	0.917
Decision Tree	0.684	0.606	0.546	0.751
Hist Gradient Boosting	0.744	0.746	0.745	0.904
Logistic Regression	**0.782**	**0.784**	**0.782**	**0.925**
MLP Classifier	0.726	0.725	0.726	0.885

Results

Model and Hyperparameters	Precision	Recall	F1 Score	AUC-ROC
Random Forest (50 Trees)	0.746	0.748	0.747	0.899
Random Forest (100 Trees)	0.761	0.761	0.761	0.913
Random Forest (200 Trees)	0.766	0.767	0.766	0.907
Support Vector Classifier	0.773	0.775	0.773	0.917
Hist Gradient Boosting (0.05 Learn Rate, No Max Depth)	0.746	0.748	0.746	0.901
Hist Gradient Boosting (0.05 Learn Rate, Max Depth 10)	0.748	0.749	0.748	0.904
Hist Gradient Boosting (0.1 Learn Rate, No Max Depth)	0.744	0.746	0.745	0.904
Hist Gradient Boosting (0.1 Learn Rate, Max Depth 10)	0.741	0.743	0.742	0.903
Hist Gradient Boosting (0.2 Learn Rate, No Max Depth)	0.737	0.739	0.738	0.898
Hist Gradient Boosting (0.2 Learn Rate, Max Depth 10)	0.738	0.740	0.738	0.898
Logistic Regression	0.782	0.784	0.782	0.925
Ensemble (Logistic Regression, SVC, and Random Forest)	**0.791**	**0.792**	**0.791**	**0.930**

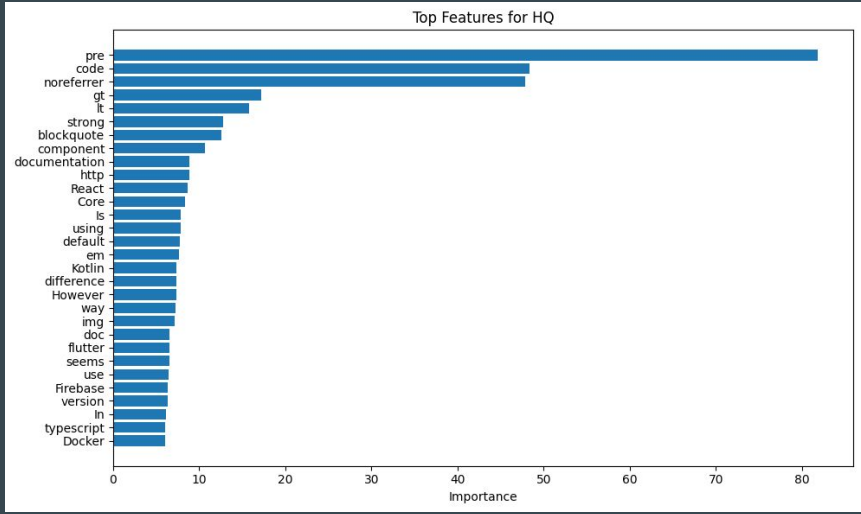
Results: Ensemble

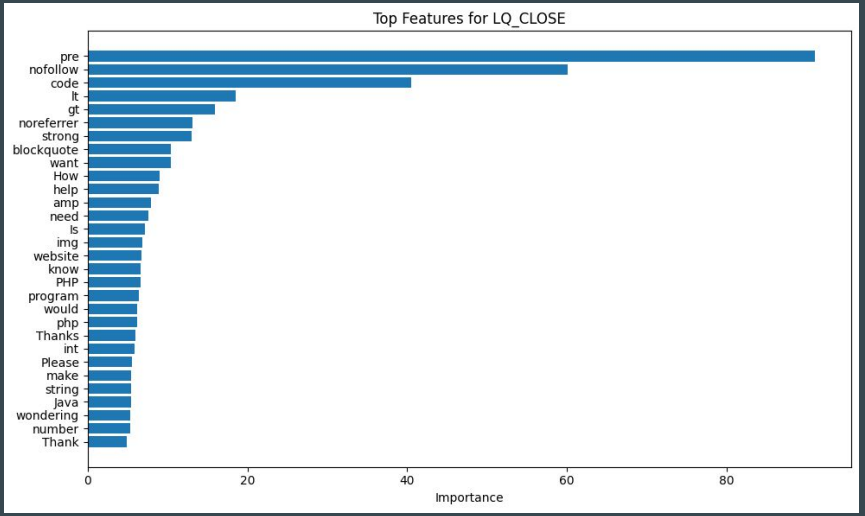
Logistic Regression + SVC + Random Forest

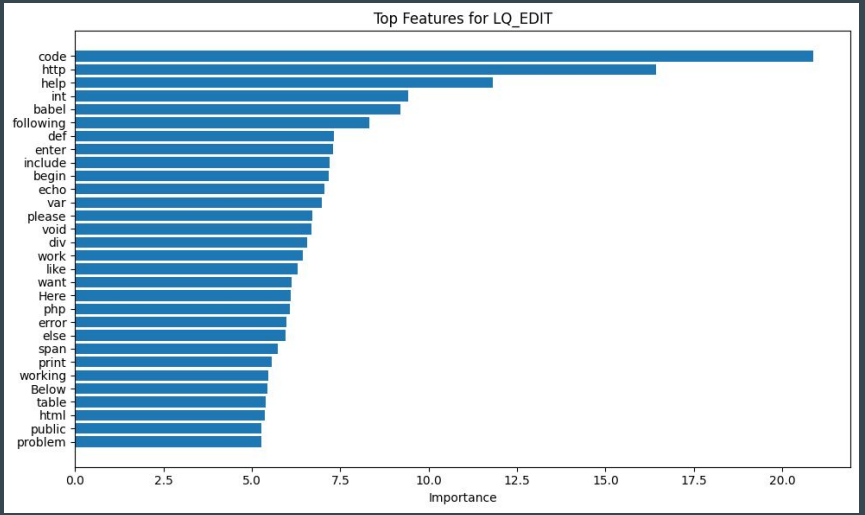


Mixed CQA	Accuracy	Avg. Precision	Avg. Recall	Avg. F1-Score
Macro Average	0.35	0.34	0.35	0.34
Weighted Average	0.35	0.34	0.35	0.34

Results: Feature Extraction







Results: Transformers

StackOverflow	Accuracy	Avg. Precision	Avg. Recall	Avg. F1-Score
Pretrained BERT + Simple Classifier	0.8075	0.8	0.81	0.81
Trained Transformers	**0.8817**	**0.89**	**0.88**	**0.88**

Mixed CQA	Accuracy	Avg. Precision	Avg. Recall	Avg. F1-Score
Pretrained BERT + Simple Classifier	0.3621	0.35	0.36	**0.31**
Trained Transformers	**0.3721**	**0.47**	**0.37**	0.30

Transformers by CQA

Pretrained BERT	Accuracy	Avg. Precision	Avg. Recall	Avg. F1-Score
AskUbuntu	0.3705	0.34	0.37	0.31
ServerFault	**0.4002**	**0.40**	**0.40**	**0.34**
SuperUser	0.3326	0.38	0.33	0.27
MetaStackExchange	0.3632	0.34	0.36	0.32
Mathematics	0.3507	0.31	0.35	0.29

Transformer Model	Accuracy	Avg. Precision	Avg. Recall	Avg. F1-Score
AskUbuntu	0.3740	0.46	0.37	0.30
ServerFault	0.3922	0.40	0.39	0.31
SuperUser	0.3074	0.44	0.31	0.38
MetaStackExchange	0.3797	0.45	0.38	0.32
Mathematics	**0.4317**	**0.53**	**0.43**	**0.35**

Limitations/Future Works

Limitations

- Only a handful of CQAs were sampled
- Some of the data might not be true to current trends due to the timeframe it was collected
- LQ_EDIT & LQ_CLOSE are similar by characteristic. Could look at determining a better way to differentiate these two classes for a more robust and accurate classifier

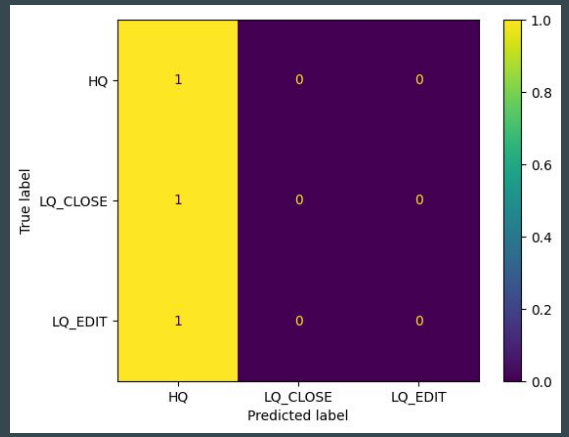
Future Works

- See how well our model performs on a CQA related to something other than computers
- Testing how well our model generalizes to CQAs offered in different languages
- See what training on the other CQA data does to the F1 and other results

How we Compare

- Compared to a dummy model our results showed that there was some overlap between Stack Overflow and the other CQAs

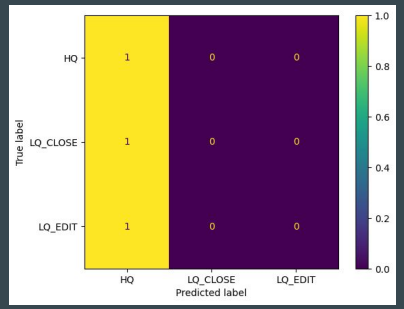
- Compared to the original paper our models didn't perform quite as well. (They were able to get a 95.6% accuracy on StackOverflow data, our highest was 88.2%)

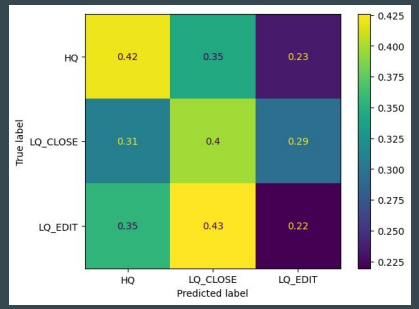


Achievements

- We were able to expand on a previous work
- We extracted features that contributed to high quality Stack Overflow posts
- We were able to train multiple models on Stack Overflow data
- We experimented with using the trained models on other CQA data to see how transferable they were

Summary





- Our models outperform untrained models.
- An ensemble of Logistic Regression, SVC, and Random Forest performs the best out of the simpler models we trained
- Our transformer performed the best overall and especially excelled on the Mathematics CQA
- When composing a question on Stack Overflow use words that are specific (like the coding language)

Questions?